

4. DCPIP is a blue dye which becomes colourless when it accepts electrons and is reduced.

A group of students extracted chloroplasts from spinach leaves. To do this they ground the spinach leaves in a mortar containing an ice-cold, isotonic solution. The resulting material was filtered and centrifuged to leave a pellet of chloroplasts. The pellet was re-suspended in an ice-cold, isotonic solution.

Tubes were prepared and treated in different ways as shown in the table. The students noted the colour of each tube at the start and after 15 minutes. The experiment was repeated three times and the same results were obtained each time.

Tube	Volume of chloroplast suspension /cm ³	Volume of DCPIP /cm ³	Volume of ice-cold isotonic solution /cm ³	Volume of distilled water /cm ³	Treatment	Colour	
						At start	After 15 minutes
1	0.5	5.0	0.0	0.0	kept in light	blue	green
2	0.5	5.0	0.0	0.0	kept in dark	blue	blue
3	0.0	5.0	0.5	0.0	kept in light	blue	blue
4	0.5	0.0	0.0	5.0	kept in light	green	green

(a) (i) Explain why ice-cold, isotonic solution was used when extracting the chloroplasts. [2]

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(ii) Explain why the students repeated the experiment three times. [1]

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(iii) Explain why the students set up tubes 3 and 4 as controls. [3]

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